

Round 8 sliver and T1 insertion: compilation harness

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2026-09-22

This harness checks the insertable fragment's typesetting. The following dependency statements are proved in Appendices A and B of `proof.md`; they are stated here only to resolve integration references. This PDF is not an independent review or a T3 certificate.

1 Notation and integration dependencies

On $(0, 1)$ use Dirichlet boundary conditions and the symmetric weight $\rho_{R,u} = R$ in $(0, u) \cup (1 - u, 1)$ and $\rho_{R,u} = 1$ elsewhere, where $R \geq 1$ and $0 < u < 1/2$. Set $D_R(u) = \lambda_2(R, u) - \lambda_1(R, u)$, $G(R, u) = RD_R(u)$, $m_R = \inf_{0 < u < 1/2} D_R(u)$, $\ell = 1/2 - u$, $\varepsilon = R^{-1/2}$, and $w = u/\varepsilon$. For $a(u) \in (\pi/2, \pi)$ defined by $\tan a = -a\ell/u$, put $\bar{D}(u) = (a(u)^2 - \pi^2/4)/u^2$.

Theorem 1.1 (Repaired large- w input). *If $R \geq 1500$ and $w \geq 2$, then $G(R, u) \geq \bar{D}(u)$. The second phase is in $(\pi/2, \pi)$ on this restricted region.*

Theorem 1.2 (Retained analytic T2). *The function $\bar{D}$ has exactly one critical point u^* . It strictly decreases on $(0, u^*)$ and strictly increases on $(u^*, 1/2)$, with endpoint limits $+\infty$ and $3\pi^2$, respectively.*

2 Insertable contribution

2.1 A whole-region analytic bound for the deep sliver

Lemma 2.1 (Pole-free phase-speed gap bound). *For every $R \geq 1$ and $0 < u < 1/2$, put $\varepsilon = R^{-1/2}$, $\ell = 1/2 - u$, and $w = u/\varepsilon$. Then*

$$G(R, u) \geq \frac{\pi^2}{2\varepsilon(w + \ell)(w + \varepsilon\ell)}. \quad (1)$$

No restriction on the second heavy-layer phase is imposed in this statement.

Proof. Let $k = \sqrt{\lambda} > 0$ and normalize the left Dirichlet solution by $y(0) = 0$, $y'(0) = 1$. At the interface its vector is $(y'(u), ky(u)) = (\cos(wk), \varepsilon \sin(wk))$. Propagation across the light interval gives

$$\begin{aligned} y'(1/2) &= \cos(wk) \cos(\ell k) - \varepsilon \sin(wk) \sin(\ell k), \\ ky(1/2) &= \varepsilon \sin(wk) \cos(\ell k) + \cos(wk) \sin(\ell k). \end{aligned}$$

Define A_ε to be the continuous real-valued argument of $\cos \theta + i\varepsilon \sin \theta$, starting at $A_\varepsilon(0) = 0$. This vector is nonzero, and

$$A'_\varepsilon(\theta) = \frac{\varepsilon}{\cos^2 \theta + \varepsilon^2 \sin^2 \theta}, \quad A_\varepsilon(\pi/2) = \pi/2, \quad A_\varepsilon(\pi) = \pi.$$

The half-interval phase is

$$\Phi(k) = \ell k + A_\varepsilon(wk), \quad 0 < \Phi'(k) \leq \ell + w/\varepsilon.$$

Its first Neumann and first Dirichlet roots satisfy, respectively,

$$\Phi(k_1) = \pi/2, \quad \Phi(k_2) = \pi.$$

They are the square roots of the first two full-interval eigenvalues. Indeed, the spatial phase increases in both layers; these two half-interval eigenfunctions are positive in their interiors. Even reflection of the Neumann solution has no interior zero, and odd reflection of the Dirichlet solution has exactly the midpoint zero. Sturm oscillation therefore identifies their full-interval indices as one and two.

Integration of the phase derivative between these actual roots yields

$$k_2 - k_1 \geq \frac{\pi}{2(\ell + w/\varepsilon)}.$$

Since $\Phi(\pi/(2w)) > \pi/2$, one has $wk_1 < \pi/2$. On this first branch, $A_\varepsilon(\theta) = \arctan(\varepsilon \tan \theta) \leq \theta$, because $\varepsilon \leq 1$. Hence

$$k_1 \geq \frac{\pi}{2(\ell + w)}.$$

Multiplying these bounds gives

$$G = \varepsilon^{-2}(k_2 - k_1)(k_2 + k_1) \geq \frac{\pi^2}{2\varepsilon(w + \ell)(w + \varepsilon\ell)}.$$

Every division above has a positive denominator. In particular, this argument includes $w < 1/2$ and does not assume $\theta_2 > \pi/2$. $\square$

Lemma 2.2 (Deep sliver, analytic replacement). *For every $R \geq 1500$ and every $0 < u \leq 2/\sqrt{R}$,*

$$G(R, u) > \frac{15\pi^2}{4} > 25, \quad G(R, u) > 3\pi^2. \quad (2)$$

Proof. Here $w \leq 2$, $\ell < 1/2$ and $\varepsilon < 1/38$, since $38^2 < 1500$. Equation (1) therefore implies

$$G > \frac{\pi^2}{2(1/38)(5/2)(2 + 1/76)} = \frac{2888}{765}\pi^2 > \frac{15}{4}\pi^2.$$

The last rational comparison is $11552 > 11475$. Also $\pi > 3$ gives $15\pi^2/4 > 135/4 > 25$, and $15/4 > 3$. This bound covers the entire continuous region, including arbitrarily small positive w , $w = 1/2$, and the infinite R tail. $\square$

2.2 T1: the infimum limit and convergence of near-minimizers

Theorem 2.3 (T1, analytic replacement). *Let u^* be the unique minimizer in Theorem 1.2, and put $M = \bar{D}(u^*)$. Then*

$$\lim_{R \rightarrow \infty} Rm_R = M.$$

More generally, let $R_j \rightarrow \infty$, $0 < u_j < 1/2$, and $\eta_j \geq 0$ satisfy

$$D_{R_j}(u_j) \leq m_{R_j} + \eta_j, \quad R_j \eta_j \rightarrow 0.$$

Then $u_j \rightarrow u^$. This includes $\eta_j = R_j^{-2}$. Moreover, $0 \leq Rm_R - M = O(R^{-1})$.*

Proof. The analytic structure in Theorem 1.2 gives

$$\bar{D}(u) \longrightarrow +\infty \quad (u \downarrow 0), \quad \bar{D}(u) \longrightarrow 3\pi^2 \quad (u \uparrow 1/2).$$

Strict increase to the right of u^* implies $M < 3\pi^2$. This inequality uses no numerical enclosure from T3.

First fix any $u \in (0, 1/2)$ and set $b = \ell/u$. For sufficiently large R one has $R \geq 1500$ and $w \geq 2$. Use the repaired large- w branch justification and Theorem 1.1, whose scope is precisely this region. Write $\theta_i = u\sqrt{R\lambda_i}$ and $a = a(u)$. On this branch, $0 < \theta_1 < \pi/2 < \theta_2 < \pi$ and $z_2 = b\varepsilon\theta_2 < \pi/4$. The odd matching relation gives

$$-\theta_2 \cot \theta_2 = \varepsilon \theta_2 \cot(b\varepsilon\theta_2) \leq 1/b = -a \cot a.$$

Since $s \mapsto -s \cot s$ is strictly increasing on $(\pi/2, \pi)$, this proves $\theta_2 \leq a$. Define

$$d_1 = \pi^2/4 - \theta_1^2, \quad d_2 = a^2 - \theta_2^2.$$

Theorem 1.1 and the deficit identity yield $0 \leq d_2 \leq d_1$. The even matching equation, with $\delta_1 = \pi/2 - \theta_1$, gives

$$0 < \delta_1 = \arctan(\varepsilon \tan(b\varepsilon\theta_1)) \leq \varepsilon \tan(\frac{\pi}{2}b\varepsilon).$$

Consequently

$$0 \leq G(R, u) - \bar{D}(u) = \frac{d_1 - d_2}{u^2} \leq \frac{\pi\varepsilon}{u^2} \tan(\frac{\pi}{2}b\varepsilon) = O_u(R^{-1}). \quad (3)$$

In particular $G(R, u^*) \rightarrow M$, so $\limsup_{R \rightarrow \infty} Rm_R \leq M$.

For every $R \geq 1500$ and every $0 < u < 1/2$, either $w \leq 2$, when Lemma 2.2 gives $G > 15\pi^2/4 > 3\pi^2 > M$, or $w \geq 2$, when Theorem 1.1 gives $G \geq \bar{D}(u) \geq M$. Taking the infimum proves $Rm_R \geq M$ and hence the asserted limit. Evaluating (3) at u^* also proves $0 \leq Rm_R - M = O(R^{-1})$.

For the stated near-minimizers,

$$R_j m_{R_j} \leq G(R_j, u_j) \leq R_j m_{R_j} + R_j \eta_j \longrightarrow M.$$

The strict sliver margin implies $w_j = u_j \sqrt{R_j} > 2$ eventually. The large- w lower bound then gives

$$M \leq \bar{D}(u_j) \leq G(R_j, u_j) \longrightarrow M.$$

Any subsequential limit of u_j in $[0, 1/2]$ is interior: a limit of zero contradicts the infinite left endpoint limit of $\bar{D}$, while a limit of $1/2$ contradicts its right endpoint limit $3\pi^2 > M$. For an interior limit, continuity and the uniqueness in Theorem 1.2 force that limit to be u^* . Thus the entire sequence converges to u^* . $\square$